

Group - A

Topics:

1. Definitions with classifications.
2. Formation
3. Order and Degree
4. Solution with method.

Ex: Variable Separable

4.1.1. Reducible to variable separable

4.2. Homogeneous Differential equation.

4.2.1. Reducible to Homogeneous Differential equation

4.2.2. Special case of Homogeneous Differential equation

4.2.3. Particular special case of Homogeneous DEs.

4.3. Linear Differential Equations.

4.3.1. Linear DE when x_0 is dependent.

4.3.2. Reducible to linear DEs / Bernoulli's DEs

4.4. Exact Differential Equation.

4.4.1. Reducible to Differential Equation

4.4.2. Integrating factors (I.F)

4.4.3. Finding Integrating Factors; Rules - 1
& Rules 2

5. Linear Differential equation with constant coefficient.

5.1. Complementary function (C.F)

5.2. Particular Integral (P.I)

Group - B

1. Bessel's Function/Equation

2. Legendre's Function/Equation

3. Method of Variation of Parameters.

4. Method of Undetermined Coefficient.

5. Lagrange's Method

6. Charpit's Method

7. Application of DEs in real world Problem.

Differential Equations

1. Definition
2. Classification
3. Ordinary Differential Equation (ODE)
4. Partial Differential Equation (PDE)
5. Formation of Differential Equation
6. Linear and Non-linear Differential Equation.
7. Order and Degree of Differential Equation.

Solution of Differential Equation

Types of Solution (General/Complete, Particular)

Method of Solution

1: Variable separable

2: Homogeneous DE

3: Linear DE

4: Exact DE

Solution : 1st order & 1st degree

Variable Separable

→ Solve $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$

Solution : Given that, $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$

or, $\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$

or, $\tan^{-1} y + c = \tan^{-1} x$

$$\text{or, } \tan^{-1} x - \tan^{-1} y = c$$

$$\text{or, } \tan^{-1} \frac{x-y}{1+xy} = c = \tan^{-1} A$$

$$\text{or, } x-y = A(1+xy)$$

$\therefore x-y = A(1+xy)$ which is the required equation.

HW : All formula of Differentiation.

Reducing to variable separable

Working rules : From pdf

$$\text{Solve : } \frac{dy}{dx} = (4x + y + 1)^2$$

Solution : Given that ,

$$\frac{dy}{dx} = (4x + y + 1)^2 \quad \text{--- (1)}$$

$$4x + y + 1 = v$$

$$\text{or, } 4 + \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{or, } \frac{dy}{dx} = \frac{dv}{dx} - 4$$

So, eqn (1) becomes,

$$\frac{dv}{dx} - 4 = v^2$$

$$\text{or, } \frac{dv}{dx} = 4 + v^2$$

$$\text{or, } \frac{dv}{4 + v^2} = dx$$

Now integrating,

$$\frac{1}{2} \tan^{-1} \frac{v}{2} = x + C$$

$$\text{or, } \frac{1}{2} \tan^{-1} \left(\frac{4x+3+1}{2} \right) = x + C$$

which is the required Equation.

Homogeneous Differential Equation

Working rule : From pdf

$$\rightarrow \text{solve } \left(x \cos \frac{y}{x} + y \sin \frac{y}{x} \right) y = \left(y \sin \frac{y}{x} - x \cos \frac{y}{x} \right) x \frac{dy}{dx}$$

Solution :

$$\frac{dy}{dx} = \frac{\left(x \cos \frac{y}{x} + y \sin \frac{y}{x} \right) y}{\left(y \sin \frac{y}{x} - x \cos \frac{y}{x} \right) x}$$

$$\text{Let } y = vx$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

so, equation (1) becomes,

$$v + x \frac{dv}{dx} = \frac{vx \left(x \cos v + v x \sin v \right)}{x \left(v x \sin v - x \cos v \right)}$$

$$\text{or, } x \frac{dv}{dx} = \frac{v \cos v + v^2 \sin v}{v \sin v - \cos v} - v$$

$$\text{or, } = \frac{v \cos v + v^2 \sin v - v^2 \sin v + v \cos v}{v \sin v - \cos v}$$

$$\text{or, } \frac{1}{x} \cdot \frac{dx}{dv} = \frac{2v \cos v}{2 \sin v - \cos v}$$

$$\text{or, } 2 \cdot \frac{1}{x} \cdot dx = \left(\frac{v \sin v - \cos v}{v \cos v} \right) dv$$

$$\text{or, } = \left(\tan v - \frac{1}{v} \right) dv$$

Now,

$$2 \ln x + \ln c = \ln \sec v - \ln v$$

$$\text{or, } \ln cx^2 = \ln \frac{\sec v}{v}$$

$$\text{or, } cx^2 = \frac{\sec v}{v} = \frac{\sec y/x}{y/x}$$

$$\text{or, } cxy = \sec y/x$$

which is required solution.

special case of HDE : Working rules: from pdf

Solve: $(2x - 2y + 5) \frac{dy}{dx} = x - y + 3$

Solution: given that,

$$(2x - 2y + 5) \frac{dy}{dx} = x - y + 3$$

or, $\frac{dy}{dx} = \frac{x - y + 3}{2x - 2y + 5}$

or, $= \frac{x - y + 3}{2(x - y) + 5} \quad \text{--- (i)}$

Let, $x - y = v$

$\therefore 1 - \frac{dy}{dx} = \frac{dv}{dx}$

or, $\frac{dy}{dx} = 1 - \frac{dv}{dx}$

So, eqn (1) becomes,

$$1 - \frac{dv}{dx} = \frac{v + 3}{2v + 5}$$

or, $\frac{dv}{dx} = 1 - \frac{v + 3}{2v + 5}$

$$= \frac{2v+5 - v-3}{2v+5}$$

$$= \frac{v+2}{2v+5}$$

$$\text{or, } \frac{dx}{dv} = \frac{2v+5}{v+2}$$

$$\text{or, } dx = \left(\frac{2v+5}{v+2} \right) dv$$

$$= \left\{ \frac{2(v+2) + 1}{v+2} \right\} dv$$

$$= \left(2 + \frac{1}{v+2} \right) dv$$

Now integrating,

$$x + c = 2v + \ln(v+2)$$

$$\text{or, } x+y = 2x-2y + \ln(x-y+2)$$

$$\text{or, } 2y-x + c = \ln(x-y+2)$$

Which is required solution.

A particular case of H.D.E :

Working Rules : From Book , page : 17 , Art-2.6 :

Solve : $\frac{dy}{dx} = \frac{2x - 3y + 5}{3x + 5y + 2}$

Solution : Given that ,

$$\frac{dy}{dx} = \frac{2x - 3y + 5}{3x + 5y + 2}$$

$$\text{or, } (3x + 5y + 2) dy = (2x - 3y + 5) dx$$

$$\text{or, } 3(xy dy + y dx) + (5y + 2) dy - (2x + 5) dx = 0$$

Now integrating

$$3xy + 5 \cdot \frac{y^2}{2} + 2y - \left(2 \cdot \frac{x^2}{2} + 5x \right) = C$$

$$\text{or, } 3xy + \frac{5}{2} y^2 - x^2 + 2y - 5x = C$$

Ans.

Linear Differential Equation :

- 1. Definition
 - 2. Working Rules
- } From PDF

Solve: $(1-x^2) \frac{dy}{dx} - xy = 1$

Solution:

Given that,

$$(1-x^2) \frac{dy}{dx} - xy = 1$$

$$\text{or, } \frac{dy}{dx} - \frac{xy}{1-x^2} = \frac{1}{1-x^2}$$

$$\text{or, } \frac{dy}{dx} + \left(\frac{-x}{1-x^2} \right) y = \frac{1}{1-x^2} \quad \text{--- (1)}$$

Now,

$$I.F = e^{\int \frac{-x}{1-x^2} dx}$$

$$= e^{\frac{1}{2} \int \frac{-2x}{1-x^2} dx}$$

$$= e^{\frac{1}{2} \ln(1-x^2)}$$

$$= e^{\ln(1-x^2)^{1/2}}$$

$$= (1-x^2)^{1/2} \quad \left[\text{As } e^{\ln x} = x \right]$$

$$= \sqrt{1-x^2}$$

Multiplying (1) by $\sqrt{1-x^2}$ we get,

$$(\sqrt{1-x^2}) \frac{dy}{dx} - \frac{x y}{1-x^2} \sqrt{1-x^2} = \frac{1}{1-x^2} \cdot \sqrt{1-x^2}$$

$$\text{or, } \frac{d}{dx} (\sqrt{1-x^2}) = \frac{1}{\sqrt{1-x^2}}$$

Now integrating.

$$\sqrt{1-x^2} \cdot y = \sin^{-1} x + C$$

$$\text{or, } y \sqrt{1-x^2} = \sin^{-1} x + C$$

Special case (y as independent variable)

Form: $\frac{dx}{dy} + p x = Q$

25.11.24

Reducible to linear Differential Eq.
*** Bernoulli's Differential Eq.

Definition: } From PDF
Working Rules: }

Solve: $\frac{dy}{dx} = x^3 y^3 - xy$

Solution: Given that,

$$\frac{dy}{dx} = x^3 y^3 - xy$$

$$\text{or, } \frac{dy}{dx} + xy = x^3 y^3 \quad \text{--- (i)}$$

Now dividing (i) by y^3 we get,

$$y^{-3} \frac{dy}{dx} + xy^{-2} = x^3 \quad \text{--- (ii)}$$

$$\text{let, } y^{-2} = v$$

$$\therefore -2y^{-3} \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{or, } y^{-3} \frac{dy}{dx} = -\frac{1}{2} \frac{dv}{dx}$$

So eqn (ii) becomes ,

$$-\frac{1}{2} \frac{dv}{dx} + xv = x^3$$

$$\text{or, } \frac{dv}{dx} + (-2x)v = -2x^3 \quad \text{--- (iii)}$$

$$\therefore \text{I.F} = e^{\int -2x dx}$$

$$= e^{-2 \cdot \frac{x^2}{2}} = e^{-x^2}$$

Now multiplying (iii) by e^{-x^2} we get,

$$e^{-x^2} \frac{dv}{dx} - 2xv \cdot e^{-x^2} = -2x^3 \cdot e^{-x^2}$$

$$\text{or, } \frac{d}{dx} (e^{-x^2} \cdot v) = -2x^3 e^{-x^2}$$

$$\text{or, Integrating, } e^{-x^2} v = \int -2x^3 \cdot e^{-x^2} dx$$

$$= \int -z e^{-z} dz$$

$$= -z \int e^{-z} dz - \int \left\{ \frac{d}{dz} (-z) \int e^{-z} dz \right\} dz$$

$$\left| \begin{array}{l} \text{let,} \\ x^2 = z \\ \therefore 2x dx = dz \end{array} \right.$$

$$-z(-e^{-z}) + \int (-e^{-z}) dz$$

$$= ze^{-z} + e^{-z} + C$$

$$\text{or, } e^{-x^2} \cdot \frac{1}{y^2} = x^2 \cdot e^{-x^2} + e^{-x^2} + C$$

$$\text{or, } \frac{1}{y^2} = x^2 + 1 + C e^{x^2}$$

Which is required solution.

Exact Differential Equation

1. Definition :

2. Necessary and Sufficient Condition ; } From PDF

3. Working Rules

$$\text{Solve : } (y^4 + 4xy^3 + 3x^2)dx + (x^4 + 4xy^3 + y + 1)dy = 0$$

Solution: Given that,

$$(y^4 + 4xy^3 + 3x^2)dx + (x^4 + 4xy^3 + y + 1)dy = 0 \quad \text{--- (1)}$$

Comparing (1) with $Mdx + Ndy = 0$ we get,

$$M = y^4 + 4xy^3 + 3x^2 \quad \text{and} \quad N = x^4 + 4xy^3 + y + 1$$

$$\therefore \frac{\partial M}{\partial y} = 4y^3 + 4x^2 \quad \text{and} \quad \frac{\partial N}{\partial x} = 4x^3 + 4y^3$$

As, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, So the given eqⁿ is exact.

Now Integrating w.r.t. x to y keeping y as constant. We get,

$$\int (y^4 + 4xy^3 + 3x^2y) dx = 4y \cdot \frac{x^2}{2} + 4y \cdot \frac{x^3}{3} + 3 \cdot \frac{x^2}{2} + C_1$$
$$= 2xy^4 + \frac{4}{3}x^3y + \frac{3}{2}x^2 + C_1$$

Again, no free terms in N is $y+1$

So integrating $(y+1)$ w.r.t. y we get,

$$\int (y+1) dy = \frac{y^2}{2} + y + C_2$$

Hence, the complete solution,

$$2xy^4 + \frac{4}{3}x^3y + \frac{3}{2}x^2 + \frac{1}{2}y^2 + y = C$$

Ans.

- Q Define Integrating factor (I.F) } From PDF
- Q. Rules for finding I.F :

Example : Q : given that,

Solve : $(3x^2y^4 + 2xy)dx + (2x^3y^3 - x^2)dy = 0$ (1)

Comparing (1) with $Mdx + Ndy = 0$, we get,

$$M = 3x^2y^4 + 2xy \quad \text{and} \quad N = 2x^3y^3 - x^2$$

$$\therefore \frac{\partial M}{\partial y} = 12x^2y^3 + 2x \quad \text{and}$$

$$\frac{\partial N}{\partial x} = 6x^2y^3 - 2x$$

As $\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ So the given eqn is not exact.

Now for exactness,

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{M} = \frac{6x^2y^3 + 4x}{3x^2y^4 + 2xy}$$

$$= \frac{2(3x^2y^3 + 2x)}{y(3x^2y^3 + 2x)}$$

$= \frac{2}{y}$, which is
function of y only.

$$\therefore I.F = e^{-\int \frac{2}{y} dy}$$

$$= e^{-2 \ln y}$$

$$= e^{\ln y^{-2}}$$

$$= y^{-2}$$

$$= \frac{1}{y^2} \left[\text{As } e^{\ln x} = x \right]$$

Now,

Multiplying (1) by $\frac{1}{y^2}$ we get,

$$\frac{1}{y^2} (3x^2y^4 + 2xy) dx + \frac{1}{y^2} (2x^3y^3 - x^2) dy = 0$$

$$\text{or, } \left(3x^2y^2 + \frac{2x}{y} \right) dx + \left(2x^3y - \frac{x^2}{y^2} \right) dy = 0$$

which is now exact.

Now,

Integrating $3x^2y^2 + \frac{2x}{y}$ w.r. to x keeping

y as constant.

We get,

$$\int \left(3x^2 y^2 + \frac{2x}{y} \right) dx = 3y^2 \cdot \frac{x^3}{3} + \frac{2}{y} \cdot \frac{x^2}{2} + C_1$$
$$= x^3 y^2 + x^2 / y + C_1$$

Again,

there is no x free term in $(2x^3 y - x^2 / y^2)$

Hence, the general solution ,

$$x^3 y^2 + x^2 / y = C \quad \underline{\text{Ans.}}$$

Linear Differential Equations with Constant Coefficients.

Definition :

- 1) linear differential equation of n th order.
 - 2) linear differential equation with constant coefficient.
- } From Book Page: 6

Note :

Theorem :

Auxiliary equation

Case - I

} From PDF

Solve:

$$\frac{d^2y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0$$

Solution: Given that,

$$\frac{d^2y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0 \quad \text{--- (1)}$$

Let, $y = e^{mx}$ be the final solution of (1),

$$\therefore \frac{dy}{dx} = m e^{mx}, \quad \frac{d^2y}{dx^2} = m^2 e^{mx}$$

So eqn (1) becomes,

$$(m^2 - 5m + 6) e^{mx} = 0$$

$$\text{But, } e^{mx} \neq 0$$

$$\therefore (m^2 - 5m + 6) e^{mx} = 0$$

$$\text{But, } e^{mx} \neq 0$$

$$\therefore m^2 - 5m + 6 = 0, \text{ which is the A.E.}$$

$$\text{or, } m^2 - 3m - 2m + 6 = 0$$

$$\text{or, } m(m-3) - 2(m-3) = 0$$

$$\text{or, } (m-2)(m-3) = 0$$

$$\therefore m = 2, 3$$

Hence the complementary function (c.f),

$$c.f = y_c =$$

Case- II (From PDF)

$$\text{Solve: } (D^2 - 2D + 1)y = 0$$

Solution: Given that,

$$(D^2 - 2D + 1)y = 0 \quad \text{--- (4)}$$

Let, $y = e^{mx}$ be the trial solution of (1).

$$\therefore Dy = me^{mx}, \quad D^2y = m^2e^{mx}$$

So eqn (1) becomes,

$$(m^2 - 2m + 1) = 0$$

$$\text{But, } e^{mx} \neq 0$$

$\therefore m^2 - 2m + 1 = 0$, which is the A.E.

$$\text{or, } (m - 1)^2 = 0$$

$$\text{or, } (m - 1)(m - 1) = 0$$

$$\therefore m = 1, 1$$

Hence the c.f. = $c_1 e^m + m c_2 e^m$ Ans.

Case-III (From PDF)

$$\text{Solve: } y_2 + y_1 + y = 0 \quad / \quad \frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = 0 \quad /$$

$$(D^2 + D + 1) y = 0$$

Solution: Given that,

$$y_2 + y_1 + y = 0 \quad \text{--- (1)}$$

Let, $y = e^{mx}$ be the trial solution of (1).

$$\therefore y_1 = m e^{mx}, \quad y_2 = m^2 e^{mx}$$

So eqn (1) becomes,

$$(m^2 + m + 1) e^{mx} = 0$$

$$\text{But, } e^{mx} \neq 0$$

$$\therefore m^2 + m + 1 = 0$$

$$\therefore m = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 1 \cdot 1}}{2 \cdot 1}$$

$$= \frac{-1 \pm \sqrt{3}}{2}$$

$$= \frac{-1 \pm i\sqrt{3}}{2}$$

$$= -1/2 \pm i \frac{\sqrt{3}}{2}$$

Hence, the complementary function,

$$y_c = e^{-\frac{1}{2}x} \left(c_1 \cos \frac{\sqrt{3}}{2} x + c_2 \sin \frac{\sqrt{3}}{2} x \right)$$

Note:

1. Particular Integral,

$$P.I. = \frac{1}{\text{l.s. of } D \text{ from excepting}} \times R.S.$$

2. If the form e^{ax} in R.S then putting $D=a$ in denominator.

3. If the form $\sin ax / \cos ax$ in R.S then putting $D^2 = -a^2$ in denominator.

Q.9. Solve: $\frac{d^3 y}{dx^3} - 3 \frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} - 2y = e^x + \cos x$

Solution: Given that,

$$\frac{d^3 y}{dx^3} - 3 \frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} - 2y = e^x + \cos x$$

or, $(D^3 - 3D^2 + 4D - 2)y = e^x + \cos x$ — (1)

Now, for C.F. consider,

$$(D^3 - 3D^2 + 4D - 2)y = 0$$
 — (2)

Let, $y = e^{mx}$ be the trial solution (2),

$$\therefore Dy = me^{mx}, D^2 y = m^2 e^{mx}, D^3 y = m^3 e^{mx}$$

or, $(m^3 - 3m^2 + 4m - 2)e^{mx} = 0$

But, $e^{mx} \neq 0$

$\therefore m^3 - 3m^2 + 4m - 2 = 0$; which is the A.E.

or, $m^3 - m^2 - 2m^2 + 2m + 2m - 2 = 0$

or, $m^2(m-1) - 2m(m-1) + 2(m-1) = 0$

$$\text{or } (m-1)(m^2-2m+2)=0$$

$$\text{or } (m-1)(m^2-2m+2)=0$$

$$\therefore m-1=0 \quad \text{or } m^2-2m+2=0$$

$$\text{or } m^2-2m+2=0$$

$$m=1$$

$$\text{or } m = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot 2}}{2 \cdot 1}$$

$$= \frac{2 \pm \sqrt{-4}}{2}$$

$$= \frac{2 \pm i2}{2}$$

Hence, the $m = 1 \pm i$

$$C.F = c_1 e^x + e^x (c_1 \cos x + c_2 \sin x)$$

Again,

$$\text{Particular Integral (P.I)} = \frac{1}{D^3 - 3D^2 + 4D - 2} (e^x + \cos x)$$

$$= \frac{1}{D^3 - 3D^2 + 4D - 2} e^x +$$

$$\frac{1}{D^3 - 3D^2 + 4D - 2} \cos x$$

As. first term case failure so multiplying by x and diff. denominator, w.r. to D

we get,
$$P.I = \frac{x}{3D^2 - 6D + 4} e^x + \frac{1}{-1^2 D - 3(-1)^2 + 4D - 2} \cos x$$

$$= \frac{x}{3 \cdot 1^2 - 6 \cdot 1 + 4} e^x + \frac{1}{3D + 8} \cos x$$

$$= x e^x + \frac{3D - 1}{9D^2 - 1} \cos x$$

$$= x e^x + \frac{3D - 1}{9(-1)^2 - 1} \cdot \cos x$$

$$= x e^x + \frac{3D - 1}{-10} \cos x$$

$$= x \cdot e^x - \frac{3}{10} D(\cos x) + \frac{1}{10} \cos x$$

$$= x e^x - \frac{3}{10}(-\sin x) + \frac{1}{10} \cos x$$

$$= x e^x + \frac{1}{10} (3 \sin x + \cos x)$$

Therefore, the complete solution,

$$y = C.F. + P.I$$

$$= c_1 e^{x_0} + e^{x_0} (c_2 \cos x_0 + c_3 \sin x_0) + x_0 e^{x_0}$$

$$\frac{1}{10} (3 \sin x_0 + \cos x_0) \text{ Ans.}$$

Bessel's Equation Bessel's Function

Q1. Define Bessel's equation.

Q2. Define Bessel's function.

Q3. Using Bessel's function prove that

$$i) J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$$

$$ii) J_{-1/2} = \sqrt{\frac{2}{\pi x}} \cos x$$

Q4. The differential equation, $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2)y = 0$

is called the Bessel's equation, where n is the order.

Q2. The solution of Bessel's equation,

$$J_n(x) = \sum_{n=0}^{\infty} \frac{(-1)^n (x/2)^{n+2n}}{\Gamma(n+1) \Gamma(n+1)} \quad \text{is}$$

called the Bessel's function.

Q3. Proof iii We know the Bessel's function,

$$J_n(x) = \sum_{n=0}^{\infty} \frac{(-1)^n (x/2)^{n+2n}}{\Gamma(n+1) \Gamma(n+1)}$$

Now putting $n = -\frac{1}{2}$, we get,

$$J_{-1/2}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n (x/2)^{-1/2} (x/2)^{2n}}{\Gamma(n+1) \Gamma(-1/2+n+1)}$$

$$= (x/2)^{-1/2} \sum_{n=0}^{\infty} \frac{(-1)^n (x/2)^{2n}}{\Gamma(n+1) \Gamma(n+1/2)}$$

$$= (2/x)^{1/2} \left[\frac{1}{\Gamma(1/2)} - \frac{(x/2)^2}{\Gamma(1) \Gamma(3/2)} + \frac{(x/2)^4}{\Gamma(2) \Gamma(5/2)} - \dots \infty \right]$$

$$= \sqrt{\frac{2}{x}} \left[\frac{1}{\sqrt{1/2}} - \frac{x^2}{2^2 \cdot \frac{1}{2} \cdot \frac{1}{2}} + \frac{x^4}{2^4 \cdot 2 \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{1/2}} - \dots \infty \right]$$

$$= \sqrt{\frac{2}{x}} \left[\frac{1}{\sqrt{\pi}} - \frac{x^2}{2 \cdot 1 \cdot \sqrt{\pi}} + \frac{x^4}{4 \cdot 3 \cdot 2 \cdot 1 \cdot \sqrt{\pi}} - \dots \infty \right]$$

$\left[\because \sqrt{\frac{1}{2}} = \sqrt{\pi} \right]$

$$= \sqrt{\frac{2}{\pi x}} \cdot \frac{1}{\sqrt{\pi}} \left[1 - \frac{x^2}{2^2} + \frac{x^4}{4^2} - \dots \infty \right]$$

$$= \sqrt{\frac{2}{\pi x}} \cos x.$$

Hence, $J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cdot \cos x$

Legendre's Equation / Function

Q1. Define Legendre's equation.

Q2. Define Legendre's generating function.

Q3. Show that, 1) $P_n(1) = 1$

$$2) P_n(-1) = (-1)^n$$

Q4. Define Rodrigues's formula. Using this evaluate

$$P_0(x), P_1(x), P_2(x), P_3(x), P_4(x)$$

Q5. Prove that, $\int_{-1}^1 [P_n(x)] dx = \begin{cases} 0, & \text{when } n \neq 0 \\ 2, & \text{when } n = 0 \end{cases}$

Q1. The differential equation,

$$(1-x^2) \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0 \quad \text{is}$$

called the Legendre's equation, where n is constant.

Q2. The equation $(1-2xh+h^2)^{-1/2} = \sum_{n=0}^{\infty} h^n P_n(x)$

is known as Legendre's generating function.

Q3. ii solution:

We know the Legendre's generating function,

$$\# \quad (1-2xh+h^2)^{-1/2} = \sum_{n=0}^{\infty} h^n P_n(x)$$

Now putting, $x=1$ we get,

$$\begin{aligned} \sum_{n=0}^{\infty} h^n P_n(1) &= (1-2xh+h^2)^{-1/2} \\ &= (1-h)^{-(-1/2)} \\ &= (1-h)^{-1} \end{aligned}$$

$$P_0(1) + h P_1(1) + h^2 P_2(1) + \dots + h^n P_n(1) +$$

..... ∞

$$= 1 + h + h^2 + \dots + h^n + \dots \infty$$

Now equating the coefficient of h^n

$$\text{We get, } P_n(1) = 1 \text{ [showned]}$$

Q4. The differential equation,

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

is called the Rodrigue's formula

Q5.

We know the Rodrigue's formula,

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

Now putting $n=2$ we get,

$$P_2(x) = \frac{1}{2^2 \cdot 2!} \cdot \frac{d^2}{dx^2} (x^2 - 1)^2$$

$$= \frac{1}{4 \cdot 2} \cdot \frac{d^2}{dx^2} (x^4 - 2x^2 + 1)$$

$$\begin{aligned}
 &= \frac{1}{8} \cdot \frac{d}{dx} \cdot (4x^3 - 4x) \\
 &= \frac{1}{8} \cdot 4 \cdot (3x^2 - 1) \\
 &= \frac{1}{2} (3x^2 - 1) \quad \underline{\text{Ans.}}
 \end{aligned}$$

Q5. Solution: We know the Rodrigues formula,

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

Now, integrating with respect to x between the limit, -1 to 1 , we get,

$$\int_{-1}^1 [P_n(x)] dx = \int_{-1}^1 \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n dx \quad (1)$$

$n \neq 0$ then, $\int_{-1}^1 [P_n(x)] dx = 0$

Again,

when $n=0$ then, $\int_{-1}^1 [P_0(x)] dx = \int_{-1}^1 \frac{1}{2^0 \cdot 0!} \cdot \frac{d^0}{dx^0} (x^2 - 1)^0 dx$

$$= \int_{-1}^1 dx$$

$$= [x]_{-1}^1$$

$$= 1 + 1 = 2$$

Hence,

$$\int_{-1}^1 [P_n(x)] dx = \begin{cases} 0, & \text{when } n \neq 0 \\ 2, & \text{when } n = 0 \end{cases}$$

Variation of Parameters.

Theorem: From PDF

$$y'' + p(x)y' + q(x)y = R(x)$$

$$\text{If } y'' + p(x)y' + q(x)y = 0$$

$$\text{then, C.F.} = C_1 y_1 + C_2 y_2$$

$\therefore$ General Solution.

$$y = u(x)y_1 + v(x)y_2$$

where,

$$u(x) = - \int \frac{y_2 \cdot R(x)}{\omega} dx,$$

$$\text{and, } v(x) = \int \frac{y_1 \cdot R(x)}{\omega}$$

$$\text{where, } \omega = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

Solve:

$$y'' + y = \operatorname{cosec} x$$

Solution: Given that,

$$y'' + y = \operatorname{cosec} x \quad \text{--- (1)}$$

Now, for C.F. consider,

$$y'' + y = 0 \quad \text{--- (2)}$$

let, $y = e^{mx}$ be the that solution of (2),

$$\therefore y' = me^{mx} \quad \text{and} \quad y'' = m^2 e^{mx}$$

So eqn (2) becomes,

$$(m^2 + 1) e^{mx} = 0$$

$$\text{But } e^{m\pi} \neq 0$$

$$\therefore m^2 + 1 = 0$$

$$\text{or } m^2 = -1$$

$$\text{or } m = \pm \sqrt{-1} \\ = \pm i$$

Hence, the C.F. = $c_1 \cos \pi + c_2 \sin \pi$.

Therefore The General solution

$$y = u(\pi) \cos \pi + v(\pi) \sin \pi \quad \text{--- (3)}$$

Now, The Wronskian

$$W = \begin{vmatrix} \cos \pi & \sin \pi \\ -\sin \pi & \cos \pi \end{vmatrix}$$

$$= \cos^2 \pi + \sin^2 \pi$$

$$= 1$$

$$u(\pi) = - \int \sin \pi \operatorname{cosec} \pi \, d\pi \\ = - \int d\pi (1 + \sin \pi) \\ = -\pi + c_1$$

Also,
$$v(x) = \int \cos x \cdot \operatorname{cosec} x \, dx$$

$$= \int \frac{\cos x}{\sin x} \, dx$$

$$= \ln(\sin x) + c_2$$

Now, putting the values of $u(x)$ and $v(x)$ in (3) we get,

$$y = (-x + c_1) \cos x + \left\{ \ln(\sin x + c_2) \right\} \sin x$$

$$= c_1 \cos x + c_2 \sin x - x \cos x + \sin x \ln(\sin x)$$

Ans.

HW:

PDF Page No. $\rightarrow$ 81

Book $\rightarrow$ 3(a), 1(a), 2

Linear Partial Differential Equation of order one

Q.1: Define Partial Differential Equation.

Note: $z = z(x, y)$ then,

$$\frac{\partial z}{\partial x} = p$$

$$\frac{\partial z}{\partial y} = q$$

$$\frac{\partial^2 z}{\partial x^2} = r$$

$$\frac{\partial^2 z}{\partial x \partial y} = s \quad \therefore \quad \frac{\partial^2 z}{\partial y^2} = t$$

Lagrange's method

The general soln of the linear partial differential equation

$$\text{eqn } Pp + Qq = R \text{ is } \phi(u, v) = 0 \quad \text{--- (1)}$$

Where ϕ is an arbitrary function and

$$u(x, y, z) = c_1 \quad \text{and} \quad v(x, y, z) = c_2$$

are solutions of eqns,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

Where, P, Q, R are functions of x, y and z .

Solve :

$$xzp + yzq = z^2$$

Soln: Given the equation,

$$xzp + yzq = z^2 \quad \text{--- (1)}$$

Comparing eqn (1) with $Pp + Qq = R$, we get,

$$P = xz$$

$$Q = yz$$

$$\text{And, } R = z^2$$

Now, By Lagrange's auxiliary eqn, we have,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\text{or, } \frac{dx}{xz} = \frac{dy}{yz} = \frac{dz}{z^2}$$

$$\text{or, } \frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z} \quad \text{--- (2)}$$

Now, From 1st and 2nd ratio we get,

integrating,

$$\ln x - \ln y + \ln c_1$$

$$\text{or, } \log(xy) = \log c_1$$

$$\therefore xy = c_1 = u \text{ (say)}$$

Again last 2nd and last two ratio we get,

$$\frac{dy}{yz} = \frac{dz}{z^2}$$

$$\text{or, } \frac{dy}{y} = \frac{dz}{z}$$

$$\text{integrating, } \log y = \log z + \log c_2$$

$$\text{or, } y/z = c_2 = v \text{ (say)}$$

Hence, the general solution,

$$\phi = (x/y, y/z) = 0$$

Ans.

H.W : Ex-2 , Ex-3 , Ex-4 , Ex-7(a) , 7(b) ,
Ex-9